Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Bengaluru, India

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Summary

DevOps Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, shipping CI/CD, IaC, and multi-account governance for enterprise financial-services and technology customers (RWS, State Street, DTCC). Deep Terraform (modules, tftest, PyTest) plus GitHub Actions, CodePipeline, CodeDeploy, and blue-green delivery on ECS. Built Terraform with automated tests for about 10 AWS services at State Street, and now automate RWS's AWS foundation in Terraform Cloud, cutting per-account setup from 3+ days to under 4 hours. 11 upstream PRs merged across axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems, 107 contests.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Bengaluru, India

- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (~90%) by replacing a legacy account-vending pipeline with Terraform Cloud automation (OIDC) for a data-centre exit and a second landing zone: 4,000+ resources across 20+ workspaces, none failing.
  - Codified Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines with IAM Identity Center, AWS Backup, and SSM Patch Manager in every workload account; every change ran in a sandbox mirror first through the same least-privilege role.
  - Built self-service network onboarding on two Transit Gateway hubs: route table within 15 minutes, firewall access by tag in about a minute. Used Kiro and Claude Code for specs and design documentation.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a **10/10 CSAT**.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC with Amazon Q.
- **MLOps Pipeline (SME Program, Jun – Aug 2025):** Built end-to-end SageMaker pipeline for medical-image classification with blue-green endpoint deployment, auto-scaling, and drift detection – provisioned via Terraform + CodePipeline with automated rollback on canary failure. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Published two more AWS samples in Terraform, on Organizations governance and a Bedrock AgentCore agent (see Publications). Co-authored an AWS Prescriptive Guidance pattern on secure PR previews and approval-gated releases for Amplify Gen 2. Delivered 2 org-wide tech talks on Terraform testing and GitOps pipeline gating. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps – AWS DevOps Agent, Agentic AI, Serverless). MLOps SME Program Graduate. Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – enabling zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for cross-service latency visibility.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code-quality analysis into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Published by AWS:** aws-samples/sample-agentcore-harness-agent-terraform (author, MIT-0) – Terraform for a Bedrock AgentCore agent (71 resources in 12 files): MCP tools behind AgentCore Gateway with a default-deny Cedar policy, PROD promotion only after an evaluation passes, CloudWatch alarms, a canary and a budget. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Projects

Ledger Sync Jan 2026
React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

orrery Oct 2026
Agent Skill, Claude Code Plugin, Node.js, ELK, SVG GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

skillcheck Jul 2026
TypeScript, Node.js, GitHub Actions, SARIF GitHub | Marketplace
◇ Conformance linter for Agent Skills: 36 rules with message-for-message parity to the skills-ref reference validator, SARIF 2.1.0 output and inline PR annotations. Ships as an npm CLI (284 downloads in the last month) and a GitHub Marketplace action.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Other Notable Projects: Bedrock Multi-Model MCP (26 text models, 156 of 156 benchmark calls), itr-agent (deterministic tax engine, 151 tests), farm-stack-aws-ecs (Docker + GitHub Actions OIDC deploy to ECR/ECS), GitScope (Chrome Web Store extension), Claude Cost Optimizer (cache-aware Claude cost model).

Open Source Contributions (11 merged upstream)

- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

IaC & Config: Terraform (modules, tftest, PyTest, remote state, S3 native locking), HCL, CloudFormation, Ansible, Packer
CI/CD & Delivery: GitHub Actions, AWS CodePipeline, CodeBuild, CodeDeploy, Blue-Green & Canary Deployments, GitOps, Policy-as-Code
AWS Platform: Organizations, SCPs, Control Tower, Transit Gateway, Direct Connect, VPN, ECS Fargate, Lambda, EventBridge, API Gateway, CloudFront, KMS, IAM, Config, Security Hub
Observability & SRE: CloudWatch (Logs, Metrics, Alarms), X-Ray, SonarQube, Drift Detection, Auto-Scaling, Incident Runbooks
Languages & Runtime: Python, Bash, TypeScript, SQL, Docker, Linux
Compliance & Governance: CCMv4, Least-Privilege IAM, Preventative Controls, Tag Policies, Cost & Usage Reporting